8 Projects (optional, for fun)

FULL

Project 9. More ideas (pick what looks fun)

Quick version

No time to read it all? This is the short version. The full guide is below.

- A list of next projects, easier to harder. Each one builds on something you already made.
- Easier: plant QR labels, a garden lights scene, a water-use diary, a dashboard on an old tablet.
- Medium: a frost reminder light, compost bin temperature, a greenhouse fan, Grafana charts, a Telegram bot.
- Harder: a time-lapse camera, a weather station, a practice copy of your website, adding your own page.
- Pick the one that makes you smile. Write one sentence: what will exist when it is done?
- Build the smallest version that works, tick it off in MY-WINS.md, then make it better.

Each idea builds on something you already made. Ask Gemini for a plan using the "design me a Home Assistant automation" prompt, or write one in this same shape: build it, check it works, then make it safer.

Easier

- **Plant name labels.** Print QR codes that open a plant's page on your site. The catalogue already supports plant links. Print and laminate a few for the beds. (Your website works at `http://localhost:8030` only on your laptop; QR codes pointing to it will only work on your home Wi-Fi unless you finish guide 04.)
- **Garden lights scene.** One button: all Govee lights to a dim warm color.
- **Water-use diary.** Home Assistant counts how long the valve ran; show it as a number card. Compare weeks.
- **Dashboard on an old tablet.** Home Assistant's dashboard in full screen, on a tablet by the back door.

Medium

- **Frost cover reminder with a light.** When frost is forecast, a light by the door turns blue until you tap "Done" on your phone.
- **Compost bin temperature.** An ESP32 with a waterproof DS18B20 probe.
- **Greenhouse fan.** A 5 V or 12 V fan on a relay (same wiring as Project 7), starting when the temperature sensor reads 80 °F / 27 °C.
- **Grafana.** A drawing program for sensor data. Home Assistant can send data to InfluxDB, and Grafana draws it. Its makers' documentation: https://grafana.com/docs/.
- **Telegram bot.** Send a photo of the bed on a schedule. The website's own camera tools do something like this on the author's server.

Harder

- **Time-lapse camera.** An old phone or a Pi camera taking a picture every hour, then `ffmpeg` joins them into a movie of a plant growing.

- **Weather station.** An outdoor BME280 and rain gauge on an ESP32 powered by a solar panel and battery. Compare it with the forecast from Project 5.
- **Compare with the real thing.** Install a second copy of your website in a clean folder, break it on purpose, and fix it. This is how people learn disaster recovery.
- **Add a page.** Follow `rust/junsgarden/src/main.rs` ("To add a page: write a function in src/pages/, add one `.route` line"). Make a page called `/hello` that says your name. Then a page that lists your last 10 sensor readings.
- **Your own sign-in gate.** 07 SSO gate in front of all your tools.

How to choose

Pick the one that makes you smile. Then do it in this order: 1. Write one sentence: what will exist when it is done? 2. List the parts. Do you own them? 3. Build the smallest version that works. 4. Tick it off in MY-WINS.md. 5. Make it better, or choose the next one.

Back to the ladder.